ChatGPT – The Super Assistant Era | BG2 Guest Interview

Podcast: BG2

AI Executive Summary

1. **Core Thesis** The OpenAI team (specifically the interviewee, an internal executive with 3.5 years of tenure) outlines a strategic pivot from scaling consumer chatbots to building autonomous AI agents capable of taking actions on behalf of users. While early growth was driven by viral adoption and foundational model releases (e.g., GPT-4), long-term success relies on maximizing user retention through friction removal, core product investment, and model improvements rather than solely optimizing revenue or acquisition. The "next billion" users will be reached not just through better models, but by evolving the product into a proactive, OS-like assistant that handles tasks (shopping, health, work) autonomously, managing trade-offs between hardware constraints (GPU availability) and feature deployment to serve diverse user segments from casual to power users.

2. **Key Arguments & Data**

3. **Growth Metrics:** ChatGPT has grown to 900 million active users (approx. 10% of the world's population), with 10% of that growth attributed to viral demo success, followed by subscriptions to shape demand and prevent capacity issues.
4. **North Star:** Retention is prioritized over revenue; specifically, long-term retention proves "durable value." There is no single dashboard metric (like WAU or DAU) that optimizes the entire goal; success is defined by whether the product helps users achieve their specific goals.
5. **Retention Data:** ChatGPT exhibits a "smile curve" in retention, where users return after gaps of weeks or months once they fully understand and integrate AI into their life processes. Key drivers for this include Search and Personalization capabilities.
6. **Pricing Strategy:** Subscriptions were initially an accidental decision to manage demand and capacity (e.g., behind a paywall during the GPT-4 breakthrough). The goal is not "unlimited electricity" pricing but maximizing access, with pilots on ads and evolving tiers for power users as technology scales.

7. **Growth Equation:** Future growth drivers are split into three equal thirds: 1) Classic friction removal (access/authentication), 2) Core product investments (search/personalization/reasoning), and 3) Pure model improvements (step changes).
8. **GPU Constraints:** GPU allocation is a "zero-sum" game that forces trade-offs between serving existing users, maintaining research momentum, and launching new features. The team prioritizes existing user experience before allocating resources to unproven breakthroughs.
9. **The Next Billion:** Requires moving from "chatbot" to "super assistant." This involves proactivity (Pulse/agents), actions (domain-specific agents like coding first, then general purpose), and removing the need for users to know *how* to delegate intelligence (OS-like affordance).

10. **User Segmentation:** Focus on both extremes: casual users (convincing them of value via simple interfaces) and power users (who teach what is possible through empirical feedback).

11. **Actionable Takeaways**

12. **For Investors/Analysts:** Monitor retention curves and token consumption per user rather than just subscriber counts; high GPU usage correlates with higher value per user in enterprise and consumer flows.
13. **For Developers:** Build domain-specific agents first (like coding tools) before expecting general-purpose autonomous action, as trust and "escape velocity" are required for broad adoption of agentic workflows.
14. **For Enterprises:** Look for AI companies capable of integrating hardware constraints into product strategy (balancing GPU allocation) and focusing on proactive features that solve "intelligence constraint problems" users face.
15. **For Educators/Students:** Prioritize "Curiosity" and high-quality writing/thinking skills as the permanent moats against AI; entrepreneurship becomes more valuable as development cycles shorten.

16. **For Product Strategy:** Adopt "Code Red" style focus periods to balance core stability (latency/reliability) with experimental research capabilities, and design interfaces that scale in complexity similar to macOS for both casual and technical users.

Full Transcript

Chat GPT originally was entirely free, and the reason for that was that it was intended to be a demo, and we were going to wind it down after a month. We then realized that the demo went viral, and people loved the demo, and it was actually a product. But we realized that to be a product, you can't take the product down every time you're at capacity. So we shipped subscriptions simply because it could shape the demand. It was a way of gracefully turning users away, and we had to turn away someone. You guys are at 900 GT active users now, and... That growth has been incredible. The next billion users, where are they gonna come from? We've got about 10% of the world coming to us now. 90% left to go, right? There's so much more opportunity. Well, Nick, so excited to have you here. Thank you for having me, Pru. You've had quite the journey from Germany to the U.S. from Brown. That's true. More recently, at Instacart, delivering groceries in 30 minutes to now delivering AGI to billions. I'm sure that was a plan all along.

Yeah, clearly total master plan. Well, tell us about your journey. How did you get to OpenAI? I know it's a fun story and you're three and a half years or so at OpenAI, how have they gone? The only through lied in how many sort of employment decisions been entirely people-based, so I don't claim any credit for joining OpenAI. We're predicting chat GPT or anything like it, but I hit up someone who I... admire a lot, who I got to know at Dropbox. Joanne, who worked here at the time, and I asked her to get off the Dolly 2 waitlist, and she told me I had an interview if I wanted to get off the waitlist. So I took the bait and got totally nerd sniped in the process, and here I am. There you go. The Dolly 2 waitlist will get you. It's a great recruiting tool. Nice, nice, nice. We should do more waitlists, probably. Yeah, yeah, yeah. Well, you know, the big...

super cycle we're in is Chad GPT. Now, I assume over a billion users on the monthly side, 900 million weekly active users that recently reported up from zero, three and a half years ago. You could have, if I imagine what the dashboard of Nick Turley looks like, it could have users, it could have paying subscribers, it could have daily active users, it could have retention

engagement. I mean, there's like 15 things, maybe all of them. What is your North Star? How do you, what are you optimizing for? What is Nick looking at on his daily dashboard? It's funny, right? Because it's such a young product. It's been, to your point, three and a half years. And this kind of question, it kind of changes as you evolve and you grow up and you ask yourself, you know, what are we really building here? And to this day, I want to build a super assistant that can actually help people achieve their goals.

And ultimately, the thing we care about is like, is our product doing that? Is it actually helping you do the thing that you're coming to the product to do? And it's so different for different people, right? Some people are trying to get healthy. Other people are trying to start a company, learn a dune topic, do their taxes. And there's all the different things that you might be doing. And the true measure of success is within that we're helping you do that. And obviously, we look at WAU in particular because we want to know if you're coming back. to the product, we look at retention, but we look at all kinds of stuff in aggregate because really there isn't like this one single thing that you can optimize for. If you were to allocate a hundred units of points to these metrics, can you distribute the hundred units across these metrics in order of importance for you to write this second? It's a good question. I care a lot about long-term retention and I would put all my points there because...

I'm really proud of the retention stats we have. But ultimately the sign of durable values, but I know people are coming back in three months because that means you're really solving their problems. And I think things like revenue, they follow from that. Versus trying to go on those things directly. And we've had a lot of success making very principled decisions on this stuff. One good example is GPT-4 used to be behind a pay wall because we couldn't serve it to everyone. And then we had GPT-4, which was a total breakthrough in... in our ability to inference it. And so we just gave it away for free and that ended up being totally revenue positive and retention positive because it just provided access to the tech. And I think when you make your decisions that way and you focus on the customer, you end up with a great product and revenue obviously follows too. Phenomenal. Yeah. Well, it shows up in the numbers. You know, I posted this chart yesterday

on the data that we have, you know, from a third party, the retention course for chat GPT are smiling.

Look at that, just like that. And that is a very rare occurrence, as we know. If you were to give us a narrative on that smile curve, why do these smile curves exist? What are you seeing in chat GPD that has people who have maybe turned off for a couple of weeks or months coming back? And why are they coming back? Look, there isn't one single thing, you know, the way that you build a retentive product is lots and lots of little things and really trying to make it better systematically. I will say that, you know, with AI and in particular ChatchaBT, I found that it takes people some time to really understand all the parts of their life they can delegate, right? And I think many users for that, it's a multi-month process for them to understand how can this thing help me and what are all the different ways that I can plug ChatchaBT into my life.

But when I think about some of the breakthroughs and levers we've had, things like search and personalization, they have helped solve those user problems because search provides way more daily value to you. It used to be that Chatchapati was a pretty work-y product. We'd say usage go down on the weekend, we'd usage go down during the summer months when a lot of people were off from work. Today, you know, we're we're mobile first the vast majority of these just mobile and we see all these personal use cases And I think search was a big investment that got us there and personalization makes Chatchity so much more relevant for you, right? because it gets to know you over time you get to know it and Those are two things that I've materially moved the way that you know people come back to the product but there's lots more to do and you know as mentioned I don't I'm not resting on our Retention stats even though we're obviously very proud Nice, nice, nice. And you know, the other thing that I got wrong about child GPT was this is two and a half years ago. And I was like, well, you know, let's look at who's going to win this consumer AI race. You typically these consumer markets are winner take most, winner take all. Look at search. Google has near 90% plus market share, three and a half, four trillion in market cap, mobile, same thing with Apple, social, same thing with Meta. And I was like, well, AI.

Meta has all the distribution. Google's got all the distribution. They've got three, four billion users. What would be a flick of a switch for them to roll out their AI? But I was wrong. That's not what happened. ChatGPT turns out, you know, you guys are at 900 million weekly active users now, and that growth has been incredible. Clearly, distribution was not enough, right? So the same question for distribution. What are the levers for us that have gotten us to the scale? Is it model quality? Is it... you know product quality is that you know features is that the experience or product improvements like memory and personalization or search like same question like what would you say drove historical growth and success we've got about 10% of the world coming to us now 90% left to go right there's so much more opportunity to reach more people and introduce them to the way that AI can benefit them right but When I look backwards, and I only say that because like the next billion users might be very different in terms of like how you engage and reach and provide value. But when I look backwards, it's been roughly a sort of one-third, one-third, one-third between sort of classic friction removal type of work, like one of the biggest moments when you look at pure impact was, you know, removing the authentication wall.

And Sam will say I told you so because I think that was his feedback from like day one It's like you can't you shouldn't have to log in these chat to PT But it's like stuff like that that you do for any product and it does matter. Yeah, some some things never change, right? And but you know then you know another third or so is you know are what I would sort of sort of core product investments And they're really typically things that we've done together between research and product So search and personalization are really good examples of that where we came together and we figure it out, not just UI UX evolution, but also how to post train these changes into the model. And it was really the moments when we came together. Another recent example is like we have these writing blocks that render when you ask about queries where you're trying to write with the model. And like putting really good craft into those experiences really matters. And our users love it. And then.

Another third of the growth has been just model improvements like step both step changes like going from GPD 3.5 back then the GPT-4 then going from GPT-4 behind a paywall to 4.0 suddenly available to everyone, right?

Yeah, but a lot of it is also the iteration that isn't splashy that doesn't warrant like, you know a name to release and we've I'm really excited about the updates we just made with 5.3 5.4 etc because that is when we like take a lot of user feedback and we, you know, methodically address it. And obviously that shows up in our retention as well. So sort of one third, one third, one third between classic friction removal and access core product investments and then pure model improvements. And so the question that I've really been waiting to ask you is how do we get the next billion? And, and, you know, talk about that a little bit. There's a lot of it seems like at least on the outside.

fog of war. If I was a consumer today in the market to pick my super assistant, you would have a couple of great options, you know, clawed out there. They're having some great traction last couple of weeks. Gemini, mega distribution, Uber distribution, and us by the leading product today, at least in user numbers, the next billion users, where are they going to come from? First of all, just to contextualize that goal, we care about two things at the end of the day. Obviously, reaching more people is really important. It's the direct manifestation of our mission to the world, where the more people we can introduce to the benefits of AI, the better that is. But we're also really excited to go deeper. And that means taking the same billion years as that, find value in chat GPT today, and actually providing more meaningful value in the world, like actually helping them achieve their goals.

not just answering questions, right? So I'll talk about how we get to more scale, but I think it's important to remember that, you know, the way the technology is evolving is, you know, we're going to go beyond pure chatbots pretty fast. I think on scale, you know, it's shocked me how many people have found value in chatBT as it works today, because I don't think delegation is a natural skill for most. And chatBT is a pretty, you know, it's a It's a power tool, right? You come to it. It doesn't tell you what it's for. You kind of have to discover it on your own, and you have to use it, and then you'll learn about this prompt that was really cool, and then maybe you're on Twitter, and you're learning about another one, or you're on Instagram, and you learn another one. But the product, it's like a raw appliance, and I think on one thing we really need to nail as we reach the next set of users is

a product that has a bit more of an affordance, because I think for most people, they're very, very busy. And everyone, I think, in the world has intelligence Constraint problems like problems that more intelligence could help with but you need to frame that to people Yeah, I still feel like we're a little bit too much like a computer terminal and it needs to feel more like software like and you know or you know an operating system of software, right? So that's one thing Another thing that gets at the same constraint is beginning to be proactive In a world where a lot of folks are too busy to delegate their problems to AI or don't quite know where to start I think being able to help you proactively is really, really important as well. But I think all of these are product evolutions that we could make on top of the current tech. And the thing that gets me particularly excited is productizing our next generation tech, our reasoning models. Because the truth is, when you look at reasoning in chatGP2 today, it's relevant to a very small group of people. It's relevant for the people who are trying to get the most out of chatGP2.

Fundamentally believe that reasoning it's transformative and if you can figure out how to productize reasoning in a way that works on people's behalf without them even knowing And that looks very much like you know the model doing long horizontal on your behalf It doesn't mean you encounter the concept. It just means it's benefiting you, right? So there's so much work to do and you know The product certainly has to evolve to to be relevant for this kind of skill. Yeah, one of the one of the things that I have been hoping for a while and, you know, Brad made a bet two years ago. When can ShadGPT help me take actions? What can ShadGPT help me be more proactive? And I think his bet expired at the end of last year, so he's very curious. When is that coming out? I'll frame that for you because, you know, with search engines in Google 20, you know, two decades ago, you got on the 10 blue links, you could have spent an hour getting the answer. You can now get the answer instantly with ShadGPT.

And it feels like the next step is actions 100% and it feels like the next step is you know when you know pulse is a great proactive product That that feels you know, it's I have a pulse that runs weekly But what I would really like is like hey Nick spoke about something and just just find me make sure I know that Nick spoke about this Or like hey this XYZ thing happened that

I cared about a lot When is when is that gonna get proactive? What is the modality going to look like? Yeah, yeah. So there's two concepts, I think. There's chatchapiti doing stuff rather than just answering. And then there's chatchapiti being proactive. And I think when you put them together, you start feeling like it feels like a super assistant because I think these things compound. On the action-taking piece, strictly speaking, chatchapiti can do stuff.

today, the action space is just very limited, right? It can search the web, which means it can use search tool or browser in the same way that a human would. It can make images. It can do all these things, right? But it doesn't have, it clearly doesn't have the same action space that a human with a computer would have. And that is what we aim to build. And if you, timing is everything on these bets, right? And I don't pretend to be great at timing either. You know, when I look at past attempts that we've made, like the Chatchapiti agent, for example, which kind of has capabilities like this. It was just slightly too early. The models weren't quite good enough to hit real escape velocity. And the problem is if you don't have escape velocity is that users don't learn to trust it. They don't even try. So when you look at a lot of things people were doing in the original version of Chatchity Agent, it was the things that happened to work, like migrating your file server into the cloud or something like that. Useful stuff, but very niche. And as the stuff gets better, We just have to get it to a point where people try to use it for real meaningful problems in their life, because then we can start hill climbing. And this has been the magic of ChatGBT, where ChatGBT upon launch was good enough to get real attempts at use cases, even if they didn't initially work. ChatGBT was a pretty bad writer originally. It was a bad software engineer, but people try to get enough value out of it that we could take those use cases and make them great. And I do think we're about to get to that point with general purpose agents where it works well enough.

that you get at least partial credit. And because you're getting partial credit, you get really good tasks back. And then the magic begins because once you have a set of use cases that you can climb the hill on, we can make them awesome. So on tasks, I think we're close, but I think even people inside of OpenAir would have had a hard time predicting exactly when this gets good. We've been excited about it for a while. On proactivity, Pulse

was a really great first step because what we wanted to build was a form factor where you're not prompting the model, like the model's prompting you. For the reasons that I described earlier, which is, you know, it's so hard for people to delegate and to figure out what their problems are. What if the eye understood your goals and things you're interested in and just could start being proactive on your behalf? Pulse is limited in the value you can provide for you because it's not connected to your life and it can't take action. So it's producing information for you.

And people love that. I love that. I've got mine running too. But I think the magic begins when you have actions and productivity because then it can begin speculatively actually detecting, hey, you just landed where you were supposed to go. I'm going to call Cap for you. Or if you're at work, it's like, hey, I proactively ran this analysis because I saw your metrics dropped. So I think these things really compound and we need to mail multiple of the building blocks to really achieve the transformation and the form factor that we hope for. As you were answering those questions, I now have 15 more questions for you. So I hope you have 15 more minutes. But OK, one by one, we'll start with what you said on actions and tasks. Got it on timing, tough to say. But is there a shape or ordinality of tasks or agents that that you think, hey, this is the kind of thing that's likely to come first whenever it does. I mean, the thing that's already come first is the domain-specific agents, right? If you look at what's happening in code, we're fully there. It's mind-bending, but we've got so many engineers who don't open their IDE ever. And for me as someone who used to code and then unfortunately got very, very busy, it's brought me back in the game. So Codex...

And you know products like it is clearly a product that has escape velocity where people are Absolutely using it for all kinds of agentic work. And if you just take what people are doing And make it work even better you kind of get all the way there You know, I won't be surprised if you see this happen for other forms of Sort of quantitative knowledge work just because it happens to have the properties that code has it's testable You know if it worked or not It's very RL friendly But The the domain specific ones already work I didn't think everyone's working for is you know general purpose agents that just kind of work for anything Yeah, and I you know that's why I think you need to to win a consumer because it's very hard to

train people into like okay It can work a deep research was a consumer product and it's really was our first agentic. Yeah thing out there But I think what consumers want is I can just ask it anything and we'll do what?

what needs to be done without any sort of retraining. We'll get there, just a matter of time. At least a psychological goal is flight bookings, hotel bookings, restaurant bookings, shopping. All this stuff, there's so many consumer problems. And those are just the type of things that you would kick off, right? The minute you have productivity, there's things you don't even think of as agentic tasks. Like you're trying to get in shape. You don't think of that as a task you would delegate. unless you have a trainer, in which case you do, but most people don't, right? But if the EI knew that, it could slowly start working in the background for you over very long periods of time and getting you, you know, here's your fitness plan. Okay, I actually signed you up for this thing. You could imagine it being quite helpful if it's aligned with your, with your long-term interest. You're going to give Osempig around for the money. We got to be careful with businesses we get into, but hopefully we can help. That'll be great. Cannot wait. Cannot wait. The second thing you said, was proactive users, and that might require us to go beyond chatbots. What's an example of a modality that might take chat GPD beyond a chatbot? So chat will always be close to my heart. It's the way we grew up, and it's an important modality to stay. I think it's less about chat and more about natural language to me, where it's the fact that you can express yourself to the machine in ways that are very natural to you, whether that's text, whether that's voice, whether or not that is...

Structured UI that is rendered by the model that is just very very powerful and that's here to stay but I think the thing essay server That's right For those who don't know that's the name of our code base I'm short for super assistant server because you know it's proof that this was always the vision and it's always the vision but you know the The thing that'll change I think is that you chat is a great way of expressing your intent It's a good way of communicating with the machine, but it's not a great output Where in many cases what you want back is an artifact. Here's your plan for your trip. Here is the analysis. Here is an outcome that I delivered for you. I just made you five bucks. Like this is what I want my AI doing for me, right? Yeah,

totally. I mean this is what people care about, right? And I think chat will always be there as the way that you sort of disambiguate your intent and you kick off the task.

But I don't think it's necessarily the final deliverable. And I think that's the way in which we can evolve. So hopefully that's a very graceful transition because I'm very lucky and it's hard earned to have a billion people coming to you weekly for a thing that they love. But I think it's a great jumping off point because we have so much unsatisfied intent from people where they're clearly trying to do something and Chatchapiti is helpful enough, but it could be so much more helpful. And I think that's where we evolve. You must be sitting on so much of this data where people are showing up to chat, GPD and attempting. As you said, three years ago, they were at least making the attempt. Yeah. So you might have at least a frequency histogram of like, hey, I heard all the things that people want to achieve with us. We do. We have like really awesome, you know, classifiers that run automatically. It's fully privacy preserving but gives us a sense of, you know, what use cases people have. And it's important, right? Because when you make a new model, we make a model update you want to know.

what use cases just got better, what use cases got worse. And that's not always, always trivial to figure out unless you have really good analytics on the system. But so much of my learning is actually qualitative, where I will just have a habit of reaching out to a fairly random set of users who just figure out what they're doing. And I've never worked on a product where three and a half years later, you're still learning every time. Because usually by that time, you know what the use cases are that your product can deliver on. But our tech is so... unusual on the fact that I keep learning about something crazy. I didn't know it was possible. Wow. That's awesome. But basically a billion users, I suspect a small fraction of them are power users who are getting maybe thousands, maybe tens of thousands of value on their 200 subscription. The vast majority is middle of the pack and a few casual users who are you know, start using ChatGPDS search maybe, or teach me about AI, or help me with my homework. What is your focus, like maybe in those constituent power users, casual users, and early users, or how server you frame it, what is our focus on for each of those three factions? Yeah, yeah. Well, first of all, I feel accountable to our entire user

base. And in fact, our non-users too, because, you know, products like ChatGPDS can have real externalities on all humans.

But when I think about sort of the way we build, it's really useful to imagine the extremes. One extreme being a user who doesn't care about AI at all, who has a busy life and needs to be convinced of the value that we can provide. Because that forces you to really nail the interface and to expose the capabilities that are hidden in the model in a way that people can actually grok. And then the other useful extreme is our power user base, because power users are the users who teach us what's possible. It's actually impossible for us to do all the product discovery on our own simply because of how empirical this technology is and how much you actually learn post launch. So building for each of those extremes can be valuable.

But our user base is incredibly diverse and people have so many different use cases and this is why, you know, I like to look at all kinds of different segmentations, not just frequency, but also, you know, what use cases are you coming to us for? But definitely huge variety in the Integra PD user base. I look up to Mac OS, for example, as an example, where it really... works for people who don't understand technology at all, it's entirely magical. But if you are a power user, you've got terminal, you've got settings, you can configure almost anything in macOS. And it's really beautifully done, where the complexity is progressively disclosed. So you can interact with it and love the simplicity of it all, but you can also have got all the knobs and developers love it, right? And so I think this is kind of the inspiration for how we want to be in chat to PT. That doesn't mean we always live up to it, but it means that building for power users is extremely important.

That's not just a property that I think is sort of aesthetically exciting. It's also really important in AI because it's the power users who show you what's possible. They are actually doing the product discovery because it would be impossible for us with such an empirical tech to do all the product discovery on our own. So the type of user who subscribes to ChatGPT Pro, who used Codex before it quite worked. who is now the strongest advocate of tools like Toget and teaching us what's possible. That is an incredibly valuable member of the community. And it might not show up in your

weekly active users. It's just one number, right? But this is exactly why there isn't a single North Star. And you really need to take these different segments very seriously. I love building for power users.

And you asked on token consumption, et cetera, it's so fascinating to see. There's people who get incredible value out of these products and watching what they do is very informative. Okay, so we're very focused on the entire user base, learn a lot from the power users. The other thing I might say is the power users right now are getting a lot of value, almost too much value. No such thing. The analog that is most common is the Uber and Lyft of the 2015 era. It took a while, but I know you were thinking about it a lot. I know you guys are thinking about pricing quite a bit. Maybe tell us a little bit about pricing. Right now pricing is pretty simple. Is there a path for folks who are getting a lot of great value to price that product differently?

and meet them where they are and the other way on the other side. And there's no world in which pricing doesn't significantly evolve when the technology is changing this quickly, right? ChatGPT originally was entirely free and the reason for that was that it was intended to be a demo and we're going to wind it down after a month. We then realized that the demo, my viral, when people love the demo, and it was actually a product, but it realized to be a product, you can't take the product down every time you're at capacity. So we shipped subscriptions simply because it could shape the demand. It was a way of gracefully turning users away when we had to turn away someone and it felt like the fairest and most equitable way of doing so is saying, hey, if you really need this product, pay a subscription fee and you got it.

But then we figured out how to meet the product stable and we had the choice of do we keep the subscription thing or do we go back to free. And we realized we had consistently more tech that we couldn't scale. GPT-4 being the first example because we had way too many free users to serve GPT-4 and we put it behind the plus plan. And so the way we stumbled into subscriptions was sort of accidental by trying to just solve for the user. And it felt like the right way at the time to provide maximal access to it. to our tech. Since then, we've had so many other breakthroughs, including test time compute where you can scale up intelligence kind of as much as you

want, more or less. And it took us in the entire industry a little bit of time to turn that into product value, but we're here now where our power is, we just want to use more and more and more intelligence.

it's possible that in the current era having unlimited plan is like having unlimited electricity plan. It just doesn't make sense because people may need a lot, a lot of electricity and they're getting a lot of value out of that. There's a reason you can't buy that, right? So obviously I want to be really thoughtful about the way that we evolve our plans and skews and subscriptions, but be incredibly surprised if it didn't change given the magnitude and profoundness of the technical breakthroughs that we've had and the product breakthroughs that follow. Yeah. And you know, relatedly, so I imagine you're going to have something for the power users. What about the other side? How do we get the casual users into the into the wheel and still monetize them? As mentioned, you know, our business model will evolve in the North Star's access. We would like to pick a We want to provide an offering that maximizes the number of people who can access our most powerful tools. I think for the longest time that has been subscriptions. Subscriptions are the downside of the fact that in many markets people don't have credit cards or they don't use credit cards to subscribe to software. And we're interested in other ways that can maximize access of the tech.

Our ads pilots are in that spirit. We really view it as a tool of bringing chat to BT and our intelligence most broadly to anyone around the world. And it is an example of how we constantly need to evolve and figure out the best way to bring the demand in line with what we are able to offer. Makes sense. Makes sense. The ads piece has been a tricky one because Sam has historically expressed reluctance about ads. you've got to maintain a lot of trust while delivering that. So I guess what changed? I think we've talked about this several times in my straight open AI. And every time it came up, we said, if we were to do ads, we'd have to be really thoughtful about the way we do it. So the first thing we did starting end of last year was to really engage the company on if we put ads in chat GPT, how should we approach it? What should the principles weigh? How do you preserve the things that are magical about chat GPT while beginning the benefits of ads, which is our ability to bring our most advanced tech to anyone, regardless of their

ability to pay? And I really love where we ended up on the principles side. On the experience side, we're very, very early.

But on the principle side, I feel really proud because it's very important that the answer of chat should be independent as an example. Respecting user privacy is very important. And there's a lot to learn from the way that tech has evolved over the last few years or really last decade. And I like that the principles are out there before we've even really gotten started. Like we're very early with our pilots. It's kind of interesting. It's obviously very... anxiously and eagerly looking at our support in bounds and data. And the most common inquiry about ads is not, you know, how do I disable ads or turn off ads, but it's like, how do I run an ad? Because the entire ecosystem is really excited to be part of the story and to figure out a way to talk to chat to be teasers. So there's a lot more to come. But I'm very eager to get this right. Yeah. Yeah, I'm sure you guys will.

Searching gears, Nick. Something you and I have spoken about a little bit is, you know, distribution and partnerships. There's a couple of big partnerships last year. Apple, Reliance with Gemini. Those are two big user bases, right? A lot of Fendi, a lot of the iOS users. Talk, tell us a little bit about how you think about partnerships for Chad GPT to meet the user base and maybe specifically on those two as well. Look, I think partnerships are a great way to bring two products together and to expose something like chat to people who might not otherwise have encountered it. The thing that I care about most when considering something like a partnership is what is the user experience and can we make it amazing? Because at the end of the day, when you look at what's going on in the market. You can get users to click on things. You get them to tap any sort of product, especially if it looks like a product they recognize, et cetera. But if the experience isn't truly awesome, people will churn or they will at least not retain in the way that we've been lucky to retain them on chat to be teeth. So for that reason, I'm super interested in paths like that.

But it needs to be great. It's a great user. We are very lucky to have a great brand and a recognizable product from many folks. And I want to make sure that anything we do is accretive to all that. Nick, you are a master of trade-offs. You must be making a lot of trade-offs right now. Tell us about

some of the trade-offs you're making. Tell us about a trade-off that you might be making that people don't appreciate from the outside. There are a lot of trade-offs indeed and for different reasons right The one I encounter a lot is trading off Delivering on the people on the use cases that exist in the product today and making them better Versus productizing you know step change technology that's gonna generate a whole another set of use cases Because when you think about how chat should be you came to be it wasn't totally open-ended product It was basically the user experience around a technical breakthrough and We couldn't have told you all the ways that people find it valuable But putting it out there was really important because it allowed us off to discover in the world to discover what you can do and then post chat should be tea We can obviously very systematically go and improve on the things that people actually want to use it for and when you're at a company in this moment where you both have such amazing traction with what exists today and the most mind-bending breakthroughs on the research side. The balance you have to strike is making the core product you have better today with all the things that matter, latency, reliability, making the use cases really great that people come to with providing access to the step change.

We try to get the balance right, but we're a small team and we don't always get it right. And for that reason, that's one of the most difficult trade-offs that I have to do it. Nick, I imagine one of the hardest trade-offs you guys make here is those GPUs that are melting between chat GPT, between codecs, research. How do you guys allocate the GPUs? That is a very good question. And I'll let you know when I figure it out. We've gotten a lot better at this. I really hope by the way to be at a point one day and I've yet to reach that point where we don't have to face this trade-off because it's really painful to have real user demand for products that you can't serve. Again, if you only ever work in software, that's entirely unusual dynamic, right, where you just, you know, you are limited by this zero sum resource out there. The marketplaces have it. But, you know, I think pure software doesn't really have the dynamic, right.

One thing we try to do, obviously, we prioritize our existing users first. We want to provide a fast, reliable product. And that is critical and table stakes. Then when you look at new capabilities, the naive business school thing to

do would be to probably look at incremental revenue per GPU or something like that. But this is where it's more an art than a science because we often have new breakthrough capabilities that are... entirely zero to one, deep research was one of those. We couldn't have told you, is there going to be demand for a consumer demand for a research product? But if you don't productize it to find out, you will never know. So this is where we have to be a little bit thoughtful on how we balance things that are no-brainers that people are really going to love with things that are brand new ideas. Then obviously on the research side, there's a reason that Mark has the job he has because a big part of His job is figuring out what research to fund and obviously GPU is a big part of that. So very nuanced topic that we're continuously getting better at. But for me, the priority is always on our users. Yeah. The other takeaway that I had is you don't have line of sight to a time when you won't have that problem. It's been so fascinating because...

We obviously have been incredibly lucky to encounter more and more users who want to use our technology. But then the value that we were able to provide for each user is going up as well. And GPU consumption correlates pretty well with that value. And when you just look at token consumption per user, especially in the enterprise too, which is a massive opportunity. you see a lot of very GPU hungry on rework flows and yes, demand keeps going up even as prices go down. This is a fascinating insight. People used to think that humans were, you know, you can't kind of make more humans. Well, it takes nine months and then 19 years. But you're saying, well, that's actually a more, less finite resource than GPUs. Yeah, I mean, on the human side, you can hire more humans. And obviously we've been doing busy doing that and bringing the best talent across functions to open it. In the world with agents, you can also get more leverage per human. You can make your humans very effective at their job to do more. But GPUs are zero-sum. And if you don't have more GPUs, you really have to figure out how do you make very, very hard trades and hate making hard trades for our users.

hence the desire to have more GPUs. But it's useful to start with the most zero-sum trade-off when you do your planning. So I think starting working backwards from GPUs is usually a pretty good idea. Yeah. We have all these external data sources for charts of users and usage and activity and

retention and all those things. What we don't have is tokens per user over time. And I bet that chart is like a sweet line going this way. I think internal is pretty good. Our internal employees is a pretty good indicator for what's about to happen. And yes, the charts are mind-boggling. Yeah, yeah, yeah. Fascinating. Okay, a couple of quick ones on the present before we go into the landscape, which is, you know, shopping. You know, we just moved into a new house. We took some photos and we were hoping that all our furniture would magically appear that tragically helped us paint.

you know, a lot of recent updates on chat GPD shopping. Tell us about it. What are you thinking on shopping as a shopping assistant? Shopping is one of those use cases that exist organically in chat GPD today and they work. You can ask chat GPD about any purchase you might be planning and get pretty excellent advice. But it's also one of those cases where the experience that exists in chat Today it's not it's not the perfect experience that you would want Because shopping is very visual for example, so you're gonna want to actually see products and images and be able to compare and contrast not just read you know walls of text And people care about the sources of you know You know where can I learn more about you know given product etc And So there's a lot of work to do to make this discovery really, really good in allowing people to use ChatGBT as an assistant to find the right product to buy. And that's where our focus lies, is making that really great and making that really great in a way that works for our retail partners as well. Because as I mentioned earlier, it is huge appetite from the ecosystem to be part of the ChatGBT journey.

And nailing the discovery piece has been the most promising focus here today. Nikon, chat GPT, you must see a breadth of information. You must see a breadth of use cases that people are doing with chat GPT. And tell us something about what does the world underestimate about chat GPT that you have maybe been surprised by or listener might be so surprised by? There's been a real change in the way that people Think of chat GPT over the last year or so where it's increasingly like a true thought partner to people. It's not just a thing that answers your question, but it's a thing that you can, it's a sparring partner that you can actually think things through with. And that shows up in all kinds of demands ranging from life advice.

relationship problem. You can actually get a lot of value of chatGPT just helping you think through how to handle it and how to talk to your partner about it all the way to a work setting where you're working on an analysis or you're like trying to figure out how to frame something or you're trying to build something and chatGPT really shows up as a second brain of sorts and I think that's qualitatively different in terms of how the mental model it occupies with people. And you see in the usage patterns in these cases that exist. And I think the more we nail things like proactivity, which we talked about earlier in tasks, et cetera, I think the more it's going to feel like a teammate in the workplace and like a super assistant at home. And I think that's going to meaningfully change the use cases that people come for. Yeah. You know, I've been the most high stakes thing I do with chat GPT is we have a new baby.

And when the baby's crying at three in the morning, chat GPT. What's going on? First of all, congrats. Second of all, I've heard this from all parents in my life. The chat GPT has become indispensable as a thought partner. And it makes sense, right? If you have a really specific scenario or you think it's a specific scenario to you, chat GPT really comes through. and can help you build confidence. And I think that's such an empowering thing, right? And imagine new parents aren't always the most confident about what is the right thing to do. And if chat should be, you can make you feel like you have agency and control. And I think that's really valuable. Yeah, it's huge. Well, thank you for.

Making chat GPT. It's literally getting me an extra sleep every day. It took a village, but that is a great metric You know that should be the north star metric is like incremental hours of sleep. That's a great one I mean you joke but like we talk about this a lot and Because spiritually that that is pretty close to what we hope we can do right this is help you reach whatever you consider self-actualization Yeah, without that sleep or joy. Yeah, or any other goal you might have. Yeah. Yeah, yeah Well, thank you for thank you to the village. We're going to switch gears and talk about the landscape. Sure, there's a lot going on on the field. You know, how would you frame chat GPT's differentiation to two people out there? There's a lot of different products out there. Look, it's the best time.

in history to be a consumer of technology because you got options and the competition is intense. And I think that's beautiful. And it's actually good for us too because if you were to pre-mortem why a company like OpenAI does not achieve its mission, it's probably focused because of the sheer number of opportunities that become possible when you approach AGI, right? And having competition and options out there, I think it forces us to focus on our customers too. and on things that really matter, which aren't always the most flashy things, right? Sometimes it's latency, reliability, the quality of the user experience. So I think it's a really good thing. I think the biggest differentiation of Judgeability is the team behind it, because we're not static, right? Anything we build will get copied, and sometimes in ways that are high-craft, sometimes in ways that are sort of checkboxes.

And it's really important to us that we evolve the category and build the super system that we've always imagined. And I think the reason that I have confidence that that's possible at a speed that outpaces the dynamic of being copied is that we have an amazing team and that amazing team across research. and engineering, and design, and all the different functions that it takes to make something amazing. And I think our unique ability has been to bring those functions together to build something that is sort of at the intersection of useful and possible right in that moment. So, you know, my best answer for you is we keep pushing forward and we hope to be expanding what people think of this product as. You know, last winter, we had, obviously, you know, what was called Code Red, Google had a great model. There was a lot of you know, talk about it, Mark Benioff switching very vocally to Gemini and us delaying ads and health agents and shopping, basically hit pause on everything, making charity better. Talk to us about that moment, both about what led to that and what was happening in that moment. Yeah. So first off, Code Reds are a tool we use to create focus. And as you can imagine, when you're in a place like open, and this is what makes us special too.

special to work here is there are so many different things going on. It's a research lab. We are pursuing many different ideas, right? And there's been these moments where we've wanted the company to come together to solve a problem across boundaries, no matter what your project might have been. And in the last year, we had one of those moments where we felt like we

need to show up for our users. We need to focus the things. focus on the basics, like reliability, performance, the way that talking to the model feels, making personalization really great, all these elements that our users care about. And I loved it because it was really an opportunity to work with a bunch of folks who I don't normally get to work with on making the product great. And we just exited the code red, which we knew we would with the launch of 5.3, which is a great model.

for the everyday user. It's great to talk to in 5.4, which is a workhorse if you're trying to do real knowledge work. Undoubtedly, we're going to continue to use the tool of a Code Red whenever we want to create focus, but I'm excited because I think chatGPT is in a great spot. Yeah, so Code Red is over now. That's correct. It's not the new normal. It's not the new normal. We want it to be a special thing, but it is a tool I suspect we will continue to use. That's great. That's great. And maybe tangibly, if you were to point out how did code red change chat GPT or maybe the ops or team operates? The thing I try to get, you know, foster with the team is focus. So we certainly more focused than we were six months ago on, you know, the things we really want to nail. And some of those things are very behind the scenes like latency, reliability, those kind of things. And some of those things are like very conservative efforts like involving chat to PT into the super assistant. And so focus is the main lasting artifact. And as you imagine, it's hard to stay focused sometimes when there's so much going on in the space. But that's the hard job. And you asked me about tradeouts earlier.

getting the team to focus on the things that really matter to users is certainly one of them that's always worth it. Yeah, you know, in the back of my mind that I asked you that question, is all the other founders that are in the arena right now, and just a reminder that, hey, Code Red is a tool for you. The wartime at Palantir, as we used to call it, is a tool. Yeah, I think, you know, every company does it differently in terms of how you get stuff done. But I think it's really valuable to have terminology that means something that signals to people it's okay to drop your other stuff and it's okay to focus on this thing together even if that wasn't your original job. So I think it works really well at a place like OpenAI. But I imagine startups would have an

equivalent. One of the things that caught everybody's imagination on our team was what Peter was doing at OpenCloud.

incredibly potent to put all the tools together. Obviously, Peter is a great builder. Congrats on bringing on Peter to the team. Tell us a little bit about what Peter is working on and when might the billions on chat GPT have something to see there? Well, first of all, I'm very excited for Peter to be here. I was excited to have another German speaker in the house. He's Austrian and I'm German. exchanging Guten Morgens. But the Open Cloud is so inspiring because it brought to life in many ways a vision that we'd had in different forms, admittedly, around this kind of AI that is fully embodied that exists across different UIs that can do stuff for you that has state that has an interaction pattern that feels a little bit more like talking to a human, because open cloud allows you to interact in a very, very natural way, where you can send many texts back and forth, and it's very curt. So there's a lot of elements of open cloud that I think we're very clarifying to folks across the industry. But I'm super excited to just learn from Peter.

bring into the company and figure out what we can do together. So there's a lot more to come. All right. So now on to the most fun section, Rapid Fire. All right. You ready? Sure. We'll start with my favorite game, which is Long Short. Pick an idea, a startup, a business, a product that you love. You think you're very bullish on? If I was starting a company today, I'm really excited about these companies that are going into companies and getting extremely hands-on and doing effectively professional services with the AI, because we've saturated all the evals and you need to get proximate to the problems. So it's those companies that I'm paying attention to. Fascinating. you know this is an example just be like hey you're going in either acquiring or going inside an operating form that has scale and a humming engine. Exactly. And making that a more efficient engine. Yeah or just like you know you're doing contracts for customers that have really hard problems and you're actually going in and committing to solving the problem. Outcomes. Yeah because like you know there's a reason I think that we've made so much progress on math and coding but not on many other domains because those are domains we are approximate to.

people who work in labs. And there's all kinds of other domains that we are not as proximate to. And if you get proximate, I think you can build something transformative. And I think this is more important now precisely because the easy problems have been solved. The obvious problems have been solved by the models. Credit where credit is due. I think Notebook LM is awesome and differentiated and helps me learn new stuff. I think it's great. It's so good. Yeah. It's so good. I think this is the example of you can innovate and you can build something totally different. It's awesome. Yeah. Yeah. Yeah. It's so good, particularly for, I found it for some more technical learning to be a very approachable way to totally learn. And it's really cool. I feel like AI, an underrated capability of AI is to just transform things into a different medium.

And I think that's so important for learning. We just launched these dynamic math blocks, which allow you to visually understand math and such at GBT. Learning is obviously a big use case for us, too. And I think just being able to transform things from text to visual, soon from visual to video, and all these different media is amazing because people have such different ways of processing information. And some people are auditory learners, some people are visual, some people are reading. So I think that's really magical and a great, great angle to take. Yeah, amazing, amazing, amazing. You know, one of the things I think about a lot is education and education for kids now in school. The world's changing so fast. I'm not sure our education system's changing that fast. What advice would you have for students who are in school now who might have to adapt faster than the system around them might adapt?

It's a really good question and something that I've thought a lot about myself and you know the I think the most important perma skill in this era is Curiosity, I think because if the machine can answer all your questions, you better have good questions And the only way to have good questions I think is to pursue the things you were actually excited about from an early age and throughout your entire life And I reflect on this because the only reason I'm here and working on this stuff is because I thought it was neat when I got, you know, nerd sniped in the interview process, right? And it's like, this is so cool. And so no matter what you're doing, I think that's an important skill is to be curious and learn to stay curious. And I think I'm

confident that if you foster that skill, you will know how to adapt to, you know, and evolving.

landscape of tools and AI's and jobs. So that would be my advice. Yeah, Curiosity has always been the porno scale. Our friend Bill Gurley wrote about it in his book, Running Down a Dream. But Curiosity. You have to check that out. Yeah. What is a job that gets more valuable, not less as AI gets better as AGI arrives? Well, I think maybe the easy answer is being an entrepreneur. Because it's the best time to build ever in terms of like being able to self-actualize your your idea Maybe one that is maybe not an obvious is I think writing actually Is very important and it's not because the AI can't write You know AI will become amazing at writing just like any other demands But because I think the skill of writing forces you to be very clear of what you have to say and even though prompt engineering is obviously going to go away and has gone away to much extent, the idea of expressing what you want to a machine requires you to be a pretty good writer and a very precise writer. So I would say that any profession that involves very clear writing and therefore thinking I think is well set up. Yeah, 100%. Honestly, I mean, this is the whole thing about Slop, right? There's just so much. That's the other thing. I think there's going to be a permanent need for high quality, trusted, authoritative content. And tools like Chatchapiti can help you discover that content. But I think the need for amazing content is also here to stay. Final question, what has been your feel the AGI moment? When did you feel it? I've had so many, honestly. And it's definitely not stopped.

A few weeks or so after I joined OpenAI GPT-24 head finish training. And I remember trying it out and it actually, it didn't impress me at all. And nor anyone else that week, because it kind of didn't work. And it's because we hadn't figured out how to post train it. And I think seeing it go from kind of, wait, is this really a thing or was GPT-23 kind of it? Wow. Actually, this is an entire step change with what felt to me at the time who didn't understand much about AI at all as like some tweaks or some a little bit of final stretch work was profoundly humbling because you can realize that you might, it might not look like we are close to really powerful, useful AI, but we probably are. And then the moment that, you know, really, you know, there was two things that GPD forwarded that felt like AGI to me.

Like one is they could do poetry, and I didn't think it was possible for an animal to do poetry. It just kind of fundamental philosophically. It just didn't feel like in scope. And then the other one was it could produce code that actually worked and compiled. And then my next moment where I stared at the ceiling just in awe was when I realized GPT-4 could just simulate an entire computer terminal. Like a full computer with commands, et cetera. And I'm like, wait, how would this be? imbued in a language model. There's been so many moments since then, honestly. Reasoning was a moment. One of the moments was when I think Mark and I were giving a demo of reasoning in front of the whole company. And this was a moment where we were still trying to find use cases that were hard enough for the reasoning team.

make it make a difference if we're way past that point we know but at the time you know I think we were having to do a puzzle in front of everyone and I think one of the moments that maybe totally feel the AGI is like we were in the middle of the demo and everyone started laughing and I was like wait what is funny and then I stared at the screen because we're showing this chain of thought as it was streaming out of the model and the model swore and said like oh damn it may have to adjust because they realized it had made a mistake in the puzzle and the fact that it did that But in particular, the fact that it did that in a way that was entirely emergent from the RL process completely blew my mind and made me feel quite humble about what else these models might be able to do. So that was one of those moments. And then most recently, watching people use Codex. Like watching people have walk around with their computer open because they don't want the task to end. Watching people who If never code in their life, make stuff and bring ideas to life, that feels like an AGM. Honestly, it's just accelerating for me and it doesn't wear off at all. And everyone has a different thing, obviously, but those are where some of mine. Yeah. You know, it's 10 years ago, there was a product called Kite. I don't know if you remember, it was for software insurance. It was like an AI coding product.

That's when I felt the hunger for personally I, and nothing happened for 10 years and then everything happened in the last 10 months. The timing thing is really hard because it's actually quite possible to predict where things will end up, I think, in terms of the kind of product and form factors you're

going to have. But to know when it happens, it's really hard for me to make statements on anything between eventually and in three months. because of all the ambiguity around, you know. Well, that's a tight enough window. You know, now in three months is a tight enough window. Three months is pretty okay. Try to stick to the three month plan more or less. Yeah. Though, you know, my team would probably tell me we don't, but I try. But yeah, anything in between three months and eventually is difficult. Yeah. Yeah, yeah. Well, thanks for doing it. You've got a lot going on. This was a total treat. We're so excited to see all the great products you release for us.

We can do anything to be of help, let us know. Awesome, thanks very much. Thanks for having me. Of course, man. This was fun. As a reminder to everybody, just our opinions not investment advice.